

A Comparative Study: TF-IDF Based Legal Judgement Recommendation System vs. Prior Approaches to Indian Legal Document Similarity

Project Exhibition – 1

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What is a recommendation model :

A recommendation system (or recommendation model) is a system designed to predict which items a user is likely to find relevant or useful, and to present those items to the user without requiring them to manually search through the entire available dataset. Recommendation systems are widely used across domains — for example, streaming platforms recommend movies or songs, and e-commerce platforms recommend products — by identifying items most similar or most relevant to a user's stated input or past behaviour.

Recommendation systems are broadly classified into three categories:

Content-Based Filtering: Recommends items based on the intrinsic features or content of the items themselves, comparing the content of a query or reference item against the content of all candidate items to find the closest matches.

Collaborative Filtering: Recommends items based on the behavior and preferences of similar users, typically relying on historical user interaction data such as ratings, clicks, or purchases.

Hybrid Systems: Combine both content-based and collaborative filtering approaches to leverage the strengths of each.

This project implements a **content-based recommendation system**. Given a user's case query or facts, the system compares the textual content of the query against the textual content of all case documents in the dataset, and recommends the cases with the highest content similarity. This approach was selected specifically because collaborative filtering requires substantial historical user-interaction data (e.g., ratings or usage patterns from many users over time) that is neither available nor practically collectible within the scope of this project. Content-based filtering, by contrast, requires only the textual content of the case documents themselves, making it both suitable and immediately deployable for this use case. This design choice is also consistent with the comparison literature reviewed in this paper (Bhattacharya et al., 2020; Mandal et al., 2021), both of which similarly adopt content/text-based similarity approaches for the same Indian legal document domain.

Abstract :

This paper presents a content-based recommendation system for Indian divorce case judgements, built using TF-IDF vectorization, cosine similarity, and a structured metadata layer (ground of divorce, verdict). To evaluate this system's design choices, we conduct a direct comparative study against two prior peer-reviewed works on Indian legal document similarity: (1) Mandal et al. (2021), which benchmarks 56 text-similarity methods including TF-IDF against expert-annotated Indian Supreme Court case pairs, and (2) Bhattacharya et al. (2020), which systematically compares network-based (citation graph) similarity against text-based similarity on the same domain. For each of these two works, we compare approach, accuracy/evaluation methodology, computational efficiency, and the specific differences and advantages of our system relative to theirs. We find that our system's design is empirically supported by both comparison papers, while offering a deployed, outcome-aware, end-user-facing application that neither prior work provides.

Keywords: Legal recommendation system, TF-IDF, cosine similarity, Indian legal NLP, comparative study, divorce case retrieval

1.Introduction :

Indian law students and junior practitioners handling matrimonial cases must sift through large volumes of prior judgements to understand how similar cases were argued and decided. This project builds a recommendation system that, given case facts or structured filters, returns similar past divorce judgements along with their verdicts.

Rather than design this system in isolation, this paper explicitly benchmarks its design against two established research works in the same problem space — Indian legal document similarity — to validate design decisions and identify genuine points of difference and advantage. The two comparison works were selected because, together, they cover the two dominant families of approach in the literature: **text-based similarity** (Mandal et al.) and **network-based similarity** (Bhattacharya et al.), allowing this paper's TF-IDF, text-based system to be assessed against both.

2. Our System: Summary

Before the comparison, we summarize the system being evaluated:

Dataset: ~56 manually curated and verified Indian divorce/matrimonial case judgements, filtered from a larger multi-domain case dataset (2016–2025)

Structured layer: each case tagged with a ground of divorce (Cruelty, Desertion, Adultery, Mutual Consent, Irretrievable Breakdown) via keyword matching

Similarity layer: text cleaned (lowercasing, stopword removal, lemmatization) → TF-IDF vectorization (scikit-learn) → cosine similarity between query and case vectors

Interface: deployed Streamlit web application offering both structured filtering and free-text similarity search, with verdict and matched keywords shown per result

2.1 Core Concepts and Techniques used :

This section briefly outlines the key technical concepts underlying the proposed system and how each is applied within the pipeline.

Content-Based Filtering: This is the overarching recommendation paradigm adopted in this work. Unlike collaborative filtering, which requires historical user interaction data to identify similar users, content-based filtering recommends items based on the intrinsic similarity of their content. In this system, a case query (either free-text facts or a structured filter) is compared directly against the textual content of stored case judgements, without any dependency on user history.

Text Preprocessing: Raw legal text contains substantial noise — inconsistent capitalization, punctuation, and grammatical variation — that must be normalized before meaningful comparison is possible. The system applies lowercasing, removal of non-alphabetic characters, stopwords removal, and lemmatization (reducing words to their dictionary root form, e.g., "granted" to "grant") to both the case dataset and any user query, ensuring both are processed identically and remain directly comparable. In addition to standard English stopwords, a domain-specific boilerplate list (e.g., "court," "petitioner," "husband," "wife," "marriage") is removed, since these terms occur in nearly every document within this dataset and therefore carry no discriminative value for similarity computation.

2.1 Core Concepts and Techniques used :

TF-IDF (Term Frequency–Inverse Document Frequency) Vectorization: TF-IDF converts each pre-processed case document into a numerical vector representation. Term Frequency captures how often a word occurs within a given document, while Inverse Document Frequency down-weights words that occur across many documents in the corpus (and are therefore less distinctive) and up-weights words that are rare and specific to fewer documents. This ensures that generic or frequently occurring legal terms do not dominate the similarity computation, while distinctive terms (e.g., "cruelty," "desertion," "irretrievable breakdown") are given proportionally greater weight. The system additionally incorporates bigrams (two-word sequences) alongside single words, allowing multi-word legal phrases to be captured as coherent units rather than as disconnected individual terms.

Cosine Similarity: Once documents are represented as vectors, cosine similarity is used to quantify how similar two vectors are by measuring the cosine of the angle between them, independent of vector magnitude (document length). A score close to 1 indicates high similarity (the vectors point in nearly the same direction), while a score close to 0 indicates little to no similarity. This metric is applied between the query vector and every candidate case vector to generate a ranked similarity score for each case.

2.1 Core Concepts and Techniques used :

Structured Metadata Filtering: In addition to free-text similarity, each case is tagged with a "ground of divorce" category (e.g., Cruelty, Desertion, Mutual Consent) derived via keyword-based classification of the case facts. This allows users to narrow the candidate case set through an explicit structured filter prior to similarity ranking, combining the precision of structured search with the flexibility of free-text similarity search.

Relevance Thresholding and Keyword-Overlap Verification: To prevent the system from recommending cases with no genuine relevance to a given query — a known limitation of naive similarity ranking, where a top-N result is always returned regardless of true relevance — two safeguards are applied. First, cases falling below a minimum cosine similarity threshold are excluded from consideration entirely, rather than being ranked and displayed regardless of relevance. Second, a candidate case must share at least two genuinely distinctive, non-boilerplate terms with the query (verified via the same TF-IDF weighting used for ranking) to be retained. Together, these mechanisms ensure that queries unrelated to the dataset's actual content (e.g., queries with no genuine legal or factual overlap with any case) correctly return no results, rather than forcing an artificially ranked but meaningless recommendation.

3. Comparison 1: vs. Mandal et al. (2021) — *Unsupervised Approaches for Measuring Textual Similarity Between Legal Court Case Reports* :

3.1 Their Approach:

Mandal et al. evaluate **56 different unsupervised text-similarity methods** — including TF-IDF, LDA topic modeling, Word2Vec, Doc2Vec, BERT, and the legal-domain-specific Law2Vec — on a dataset of Indian Supreme Court case reports. Each method's output similarity score is compared against a **gold-standard similarity score** produced by legal experts manually rating document pairs, using **Pearson correlation** as the evaluation metric.

3.2 Approach Difference:

Their work is a **research comparison study** — it does not build a single deployed system, but rather evaluates many candidate methods against each other in a controlled experimental setting. Our system, by contrast, is a **single deployed method** (TF-IDF + cosine similarity) embedded in an end-to-end, user-facing application with a structured filtering layer on top, which their study does not include.

3. Comparison 1: vs. Mandal et al. (2021) — *Unsupervised Approaches for Measuring Textual Similarity Between Legal Court Case Reports* :

3.3 Accuracy Comparison:

Mandal et al. report Pearson correlation scores between each method's similarity output and expert-annotated gold-standard scores across their 56 methods. **A directly equivalent accuracy figure could not be produced for our system**, since this would require recruiting legal experts to manually annotate similarity scores for a comparable set of document pairs — outside the scope achievable for this project. In place of this, our system's relevance was assessed **qualitatively**: for representative test queries across each ground-of-divorce category, the top-ranked results were manually checked by the project team for topical relevance. This is a genuine limitation relative to Mandal et al.'s rigorous evaluation, stated plainly rather than substituted with an invented number.

3.4 Efficiency Comparison

This is where the comparison is most favorable to our approach. Mandal et al.'s 56-method sweep includes computationally expensive transformer-based methods (BERT), which require significantly more compute (ideally GPU acceleration) both to train/fine-tune and to run inference, compared to TF-IDF, which requires neither training nor specialized hardware. **Critically, Mandal et al.'s own reported finding is that simpler, traditional bag-of-words methods (including TF-IDF) perform competitively with — and in several configurations outperform — the more complex embedding-based methods for this specific Indian legal text domain.** This is a directly citable empirical validation of our system's choice of TF-IDF: it is not merely a simpler fallback, but a method with documented competitive accuracy at substantially lower computational cost.

3. Comparison 1: vs. Mandal et al. (2021) — *Unsupervised Approaches for Measuring Textual Similarity Between Legal Court Case Reports* :

3.5 What Our System Does Differently / Better:

- **Lower Computational Overhead** for comparable relevance, per their own findings.
- **A deployed, usable interface** – theirs a benchmarking study with no end-user application.
- **Verdict/outcome shown alongside recommendations** – their evaluation is purely about similarity scoring, not about presenting actionable case outcomes to a user.
- **Structured filtering (ground of divorce) combined with similarity search** – their work evaluates text-similarity methods in isolation, without a structured metadata layer.

4. Comparison 2: vs. Bhattacharya et al. (2020) — *Methods for Computing Legal Document Similarity: A Comparative Study*:

4.1 Their Approach:

Bhattacharya et al. systematically compare **network-based similarity** (built from the citation graph between case documents — i.e., which cases cite which) against **text-based similarity** methods, on Indian Supreme Court cases and their associated statute citation network. They also propose and test new network-embedding-based and text-based methods not previously evaluated in this comparison.

4.2 Approach Difference :

Their central approach relies on constructing and analyzing a **citation network** — this requires case documents to have rich, resolvable citation links to other cases, which must be extracted and structured beforehand. Our system uses **no citation network at all**; it relies purely on the textual content of case facts. This is a fundamental architectural difference: their strongest results come from combining network and text signals, while our system is text-only by design, prioritizing simplicity and independence from citation-data availability.

4.3 Accuracy Comparison:

As with Mandal et al., Bhattacharya et al. use Pearson correlation against expert-annotated similarity gold standards for evaluation. The same limitation applies: our system's evaluation is qualitative (manual relevance-checking), not a statistically validated correlation score, since we did not have access to expert legal annotators for gold-standard scoring. This is stated here as an explicit, acknowledged gap rather than papered over.

4. Comparison 2: vs. Bhattacharya et al. (2020) — *Methods for Computing Legal Document Similarity: A Comparative Study*:

4.4 Efficiency Comparison:

Building and maintaining a citation network — as required by Bhattacharya et al.'s approach — requires citation extraction, graph construction, and (for their best-performing methods) network embedding computation, all of which add meaningful preprocessing and maintenance overhead. Bhattacharya et al. themselves, along with related work in this space, note that **Indian legal citation networks are often sparse and unreliable when used in isolation** — a structural limitation of the network-based approach itself. Our system, by avoiding network-based methods entirely, is immune to this sparsity problem and requires no citation-graph infrastructure, making it substantially simpler to build and deploy for a bounded, smaller-scale dataset such as ours.

4.5 What Our System Does Differently / Better:

- **No dependency on citation network availability or density** – avoids the sparsity problem their own paper identifies as a limitation of network-based methods.
- **Simpler pipeline** – No graph construction or network embedding step required.
- **Deployed applications Vs. research comparison** – again, theirs is an experimental study; ours is a working tool.
- **Domain – bounded and outcome aware** – our system is purpose-built for divorce cases specifically, with verdicts shown directly, rather than general-purpose similarity scoring across all Supreme Court cases.

Core approach	Compares 56 text-similarity methods (TF-IDF, LDA, Word2Vec, Doc2Vec, BERT, Law2Vec)	Compares network-based vs. text-based similarity	TF-IDF + cosine similarity + structured ground-of-divorce tagging
Type of work	Research benchmarking study	Research benchmarking study	Deployed end-user application
Dataset	Indian Supreme Court cases (large-scale)	Indian Supreme Court cases + statute citation network	~56 curated, manually verified divorce cases
Evaluation	Pearson correlation vs. expert gold standard	Pearson correlation vs. expert gold standard	Qualitative manual relevance check
Computational cost	High (includes BERT, GPU-favorable)	High (citation graph construction + embeddings)	Low (TF-IDF, no training, runs on standard hardware)
Citation network dependency	No	Yes (central to approach)	No
Outcome/verdict shown to user	No	No	Yes
Deployed, user-facing system	No	No	Yes
Domain bounding	General Supreme Court corpus	General Supreme Court corpus	Bounded to divorce/matrimonial law

Method	Metric used	Reported score	Notes	
Bhattacharya et al. — Co-citation	Pearson correlation	0.205	Weakest network-based method tested	
Bhattacharya et al. — Dispersion	Pearson correlation	0.246	Network-based	
Bhattacharya et al. — Bibliographic Coupling	Pearson correlation	0.443	Network-based	
Bhattacharya et al. — Node2Vec	Pearson correlation	0.487	Best network-based method	
Bhattacharya et al. — Paragraph Links	Pearson correlation	0.330	Text-based	
Bhattacharya et al. — Thematic Similarity (avg)	Pearson correlation	0.523	Text-based, segment-wise	
Bhattacharya et al. — Thematic Similarity (max)	Pearson correlation	0.599	Text-based, segment-wise	
Bhattacharya et al. — FullText/Doc2Vec	Pearson correlation	0.605	Best single text-based method	
Bhattacharya et al. — Best combined (FullText + Paragraph Links)	Pearson correlation	0.626	Best overall in their study	

6. Discussion :

Both comparison papers validate distinct aspects of our system's design. Mandal et al.'s finding — that simple bag-of-words methods including TF-IDF are competitive with far more complex embedding methods on Indian legal text — directly supports our choice of TF-IDF as neither naive nor merely convenient, but empirically justified. Bhattacharya et al.'s identification of citation-network sparsity as a limitation of network-based methods supports our decision to avoid network-based similarity entirely, particularly given our smaller, bounded dataset where a citation graph would likely be even sparser than in their large-scale Supreme Court corpus.

The clearest and most consistent gap between our system and both comparison papers is that **neither is a deployed, user-facing application**, and neither surfaces case **outcomes** to the end user — both papers are focused purely on the underlying similarity-scoring problem as a research question. Our system's contribution is applying validated similarity-scoring principles from this literature into an actual usable tool, purpose-built for a specific legal domain and audience (law students), with outcomes shown directly alongside recommendations.

The clearest and most honest limitation of our system, relative to both, is evaluation rigor: without expert-annotated gold-standard similarity scores, we cannot report a statistically comparable accuracy figure, and rely instead on qualitative relevance verification.

7. Conclusion and Future Scope :

This comparative study positions our TF-IDF-based divorce case recommendation system against two established approaches to Indian legal document similarity, finding that our core methodological choices are empirically supported by both works' own findings, while our system offers a deployed, outcome-aware application that neither comparison work provides. Future work should prioritize closing the evaluation gap identified in Sections 3.3 and 4.3 — specifically, constructing a small expert-annotated similarity gold standard to enable a direct, statistically valid accuracy comparison against both Mandal et al. and Bhattacharya et al. in future iterations of this project.

References :

1. Mandal, A., Ghosh, K., Ghosh, S., Mandal, S. "Unsupervised Approaches for Measuring Textual Similarity Between Legal Court Case Reports." *Artificial Intelligence and Law*, 29(3), 417–451, 2021.
2. Bhattacharya, P., Ghosh, K., Pal, A., Ghosh, S. "Methods for Computing Legal Document Similarity: A Comparative Study." arXiv:2004.12307, 2020.

THE END
